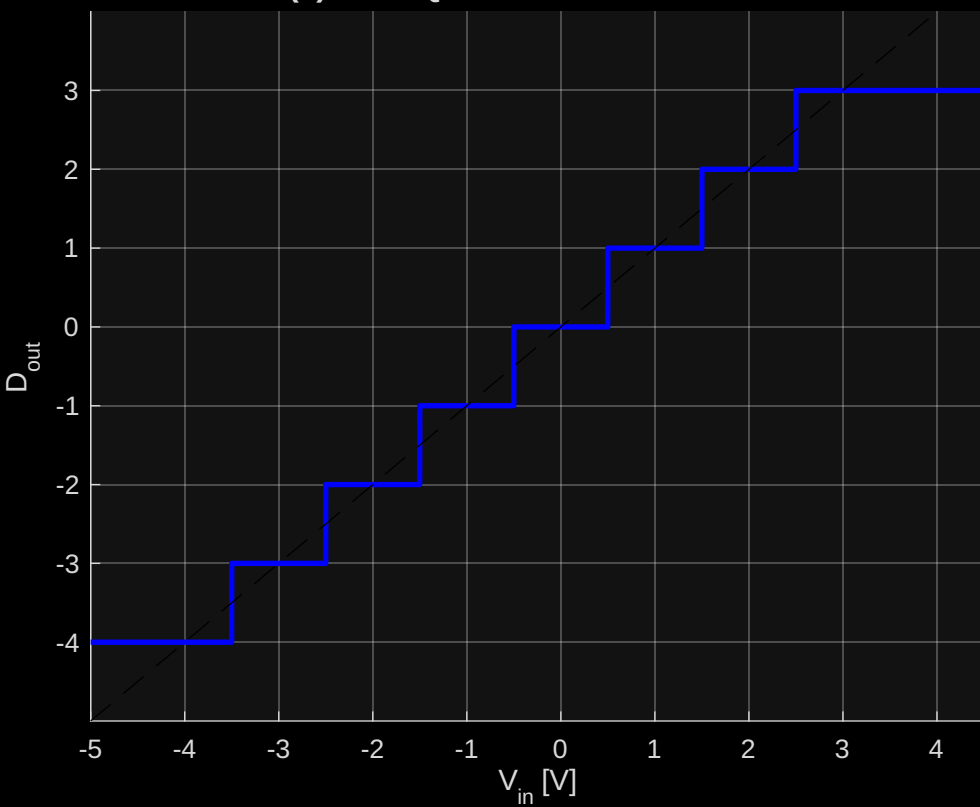
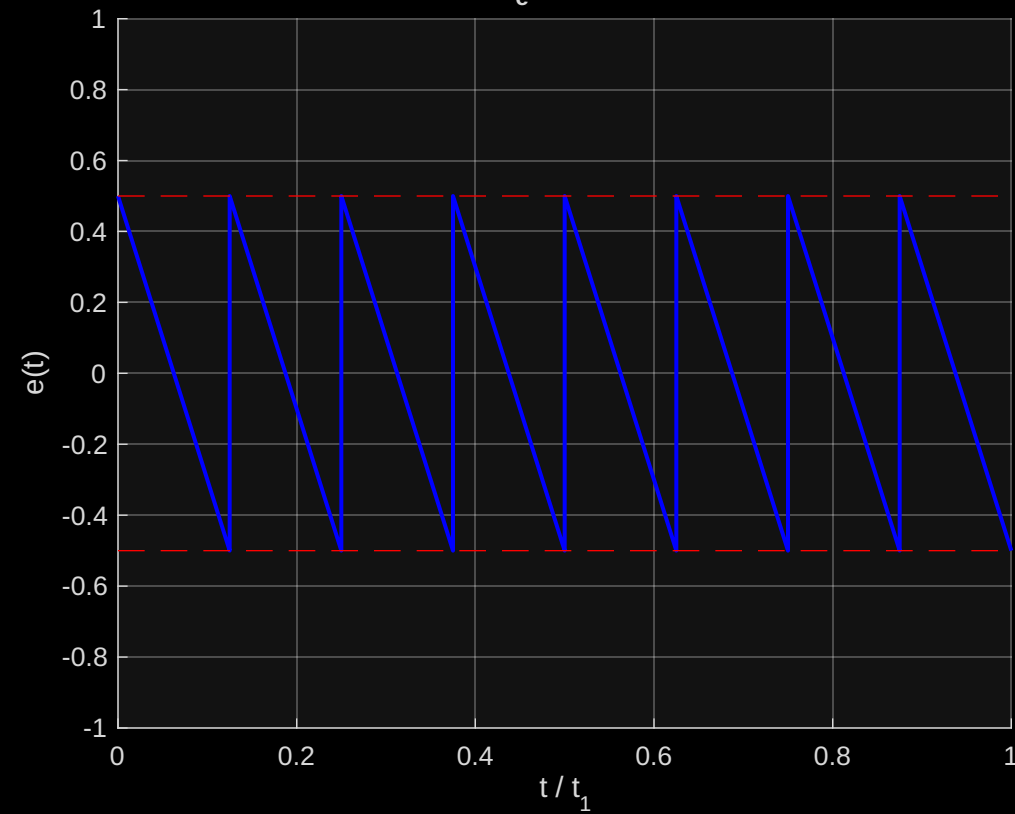


Problem 3: Quantization Noise Power for Ramp Input

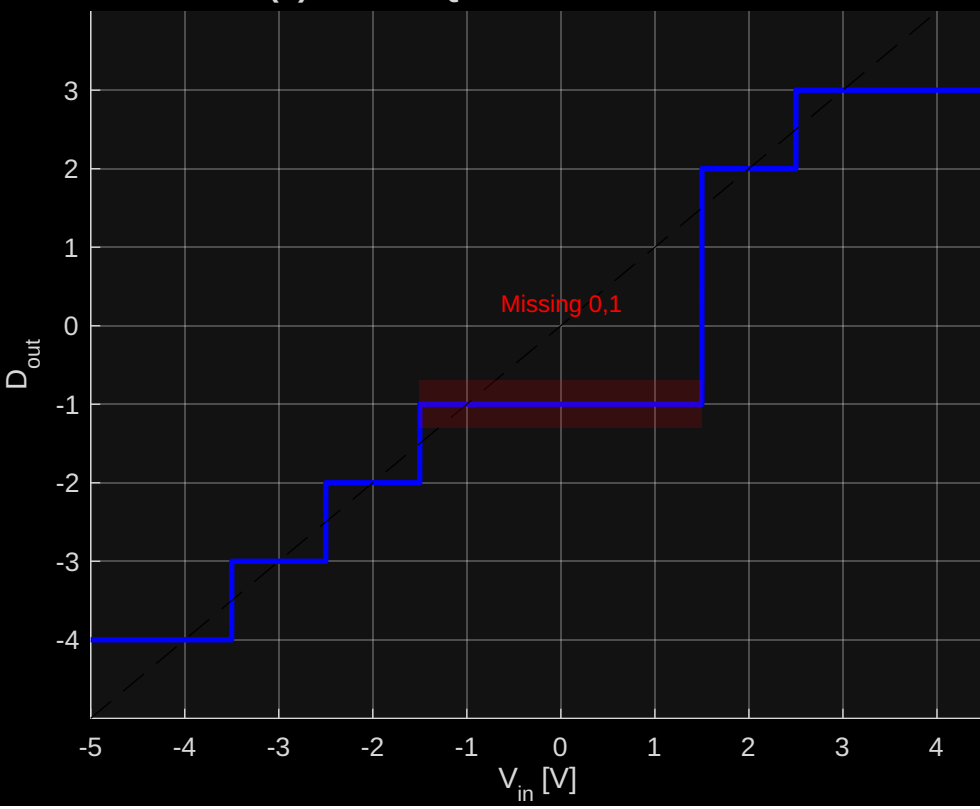
(a) Ideal Quantizer Characteristic



(a) Error: $P_e = \Delta^2/12 = 0.0833$



(b) Broken Quantizer Characteristic



(b) Error: $P_e = 17\Delta^2/24 = 0.7083$

